

# Kasi Vandanapu

Senior Backend / AI Engineer | AI Systems & Data Platforms

Senior Backend / AI Engineer with 5+ years of experience building production AI systems and data platforms in Python. Specialize in agentic workflows, Text2SQL evaluation, and the infrastructure that makes them reliable and cost-efficient — delivering **50%** lower LLM inference cost, **60%** faster pipelines, and **30%** faster Spark jobs.

## EXPERIENCE

**Lead Python Developer**  
Virtusa · Consulting for Citi Bank

Jun 2024 – Present  
Tampa, FL

- Lead the Python team for **Maphub** (Citi's internal data management platform); own **Balance Sheet Management** and **Capital Data** use cases end-to-end.
- Built LLM-powered agentic workflows with **LangGraph**/LangChain to generate data models, UI configurations, and dashboards from available data and visualization context.
- Developed a **Text2SQL benchmark framework** evaluating LLM performance on enterprise schemas using EX and QAS metrics.
- Cut prompt token usage and LLM inference cost **50%** via schema-graph pruning, DDL compression, and relevance-ranked table retrieval, holding execution accuracy constant.
- Designed **FastAPI** microservices to extract, validate, transform, and load data into MongoDB and Oracle, using **Polars** and Pandas to process 100K rows in under 10 seconds.
- Integrated real-time ML with **AWS AutoGluon** — UI-driven training, versioning, and deployment of live prediction models in Maphub Data Studio.

**Specialist Programmer**  
Infosys

May 2019 – Dec 2021  
Hyderabad, India

- Built ETL pipelines with **PySpark**, Delta Lake, and Airflow on Databricks for **Levi's**, **Kraft Heinz**, **HCSC**.
- Engineered a **GCP** streaming pipeline on Pub/Sub and Cloud Run; tuned Spark jobs to cut runtime **30%**.

**Software Developer, Intern**  
Vega Innovations · For Health Edge Inc.

Feb 2019 – May 2019  
Bangalore, India

- Developed and tested microservice applications using **Spring Boot**, Hibernate, and Apache Maven.

## FEATURED PROJECTS

**In-Browser Analytics with DuckDB** · Technical Writing

2026

Published deep-dive on client-side analytics — DuckDB-WASM, Parquet, Mosaic for serverless crossfiltered dashboards.

**Wander Finds** · Gemini AI Hackathon

2024

AI travel app generating personalized recommendations from Gemini multimodal and location-aware Maps context.

**Text2SQL Benchmark** · Open-Source Framework

2024–25

Open-source LLM evaluation framework — composite EX + QAS scoring, intent-aware column selection.

## SKILLS

**AI / LLM**  
LangGraph · LangChain · Agentic AI · Text2SQL · LLM Evaluation (EX/QAS) · RAG · AutoGluon · XGBoost

**LANGUAGES**  
Python · SQL · JavaScript · Bash

**BACKEND**  
FastAPI · REST · Microservices

**DATA**  
PySpark · Databricks · Delta Lake · Airflow

**ANALYTICS**  
DuckDB · Apache Arrow · Mosaic

**CLOUD**  
AWS · GCP · Kubernetes

**DATABASES**  
Oracle · MongoDB · MySQL · DuckDB

## EDUCATION

**M.S. Computer Science**  
Northeastern University

Jan 2022 – Dec 2023 · San Francisco, CA

Teaching Assistant — DBMS & HCI

**B.Tech Computer Science**  
Amrita School of Engineering

Aug 2015 – May 2019 · Bangalore, India

## CERTIFICATIONS

**PCAP™ — Certified Associate Python Programmer**  
Python Institute

**Smart Analytics, ML & AI on GCP**  
Google Cloud

## RESEARCH

**Research Assistant**  
Northeastern University

May 2022 – Dec 2023 · In-browser analytics, Observable

## PUBLICATION

“Hadoop and NLP Based Analysis on Kisan Call Center (KCC) Data” — Intl. Conference on Advances in Computing, Communication & Informatics, 2018.